ABO Blood Groupings

Human blood type is determined by 3 genes: A, B or O. As usual, these genes occur in pairs. A and B are both dominant and so will always show their effect if present. They are both dominant to O, which is a recessive gene.

Fill in the table

Genes	OO	AA	AO	BB	BO	AB	Genotype
Blood Type							Phenotype

1. What genes could a type-A person have?
2. What genes could a type-B person have?
3. If a type-A mother and a type-B father have a type-O child, what does this tell us about the genotypes of parents? Use a punnett square to explain your answer.
4. What are the possible blood types of the offspring of a type-O person and a type-AB person?
5. Two groups of parents think that the hospital may have accidently switched their new babies. Mrs Jones has O-type blood and Mr Jones has AB. The baby labelled “Jones” has B-type. Mr and Mrs Smith both have B-type blood. The baby labelled “Smith” has O-type blood. Do you think there was a mistake? Show your working using punnett squares.
6. In the 1950’s, a young woman sued film star/director Charlie Chaplin for parental support of her illegitimate child. Charlie Chapin’s blood type was already on record as type AB. The mother of the child had type A and her son had type O blood.
    a. What are the possible blood types of the mother?
    b. Complete punnett squares for all possible crosses between Charlie and the mother.
    c. The judge ruled in favour of the mother and ordered Charlie Chaplin to pay child support costs of the child. Was the judge correct in his decision based on blood typing evidence? Explain why or why not using your punnett squares from part c.

More Incomplete Dominance

1. The genes for flower colour in Four o’clock flowers are incompletely dominant.
    a. Predict the genotypes and phenotypes of the offspring when a white flower and a red flower are crossed.
    b. Two flowers from the F_1 generation are crossed. What are the expected frequencies of genotypes and phenotypes from this cross?

2. In humans straight hair and curly hair are incompletely dominant traits, that result in hybrids with wavy hair.
    a. A curly haired female marries a wavy haired male. What are the possible genotypes for their offspring?
    b. What are the chances of having a curly haired offspring?

Pedigrees

3. Dimples are an inherited characteristic and are dominant over no dimples.
    a. Draw a pedigree to show a family where the father has dimples and the mother doesn't. All their four children, three male and a female, have dimples. The eldest child marries a woman with dimples and they have two children, a boy with dimples and a girl without.
    b. What are the genotypes of all the people in the family tree?